8-5 Network Transfer Speed

Point!

Network Transfer Speed

(1) Transfer speed : The ([1]number of bits) of data that can be transferred per second. The unit is ([2]bps).
1 kbps = ([3]1,000 bps), 1 Mbps = ([4] 1,000 kbps), 1 Gbps = ([5]1,000 Mbps)

(2) The relationship between transfer speed, transfer time, and data volume

[1] Transfer speed [bps] = $\left([6]\ \frac{\text{Data volume [bits]}}{\text{Transmission time [seconds]}}\right)$

[2] Transfer time [seconds] = $\left([7]\ \frac{\text{Data volume [bits]}}{\text{Data transfer speed [bps]}}\right)$

[3] Data volume [bits] = ([8]Transfer speed [bps] × Transfer time [seconds])

(3) ([9]Transfer efficiency) : The ratio at which a certain communication speed can be achieved.
Actual transfer speed = ([10]Communication speed [bps] × Transfer efficiency)

Warm Up

Answer the following questions. In these questions, assume that 1 KB = 1,000 B, 1 MB = 1,000 KB, and 1 GB = 1,000 MB.

(1) Calculate the time required to transfer 20 MB of data at a communication speed of 40 Mbps. Assume that the transfer efficiency is 100% and disregard any data other than the data volume.

(2) Calculate the time required to transfer 1 GB of data with a communication speed of 100 Mbps. Assume that the transfer efficiency is 80% and disregard any data other than the data volume.

Explanation

(1) Transfer time [seconds] = $\frac{\text{Data volume [bits]}}{\text{Transfer speed [bps]}}$

1 B = 8 bits, 1 KB = 1,000B, 1 MB = 1,000 KB, therefore,
20 [MB] = 20 × 1,000 × 1,000 × 8 [bits]
Also, 40 [Mbps] = 40 × 1,000 × 1,000 [bps] — Convert units
Therefore, the calculation of transfer time is as follows:

$\frac{20 \times 1{,}000 \times 1{,}000 \times 8\ [\text{bits}]}{40 \times 1{,}000 \times 1{,}000\ [\text{bps}]} = 4\ [\text{seconds}]$ 4 seconds

(2) 1 [GB] = 1 × 1,000 × 1,000 × 1,000 × 8 [bits]
Moreover, since the transfer efficiency is 80%, the actual transfer speed is 100 [Mbps] × 0.8 = 80 [Mbps]
80 [Mbps] = 80 × 1,000 × 1,000 [bps], therefore,
the calculation of transfer time is as follows:

$\frac{1 \times 1{,}000 \times 1{,}000 \times 1{,}000 \times 8\ [\text{bits}]}{80 \times 1{,}000 \times 1{,}000\ [\text{bps}]} = 100\ [\text{seconds}]$ 100 seconds

Try

Answer the following questions. In these questions, assume that 1 KB = 1,000 B, 1 MB = 1,000 KB, and 1 GB = 1,000 MB.

(1) Bps is a unit used to measure data communication speed. Choose the appropriate explanation for 20 Mbps from the options **A** to **D** below, and answer using the letter.
- **A** Capable of transferring data at 20 megabits per minute.
- **B** Capable of transferring data at 20 megabytes per minute.
- **C** Capable of transferring data at 20 megabits per second.
- **D** Capable of transferring data at 20 megabytes per second.

(2) Calculate the maximum data volume, in bytes, that can be transmitted in one minute at a communication speed of 30 bps.

(3) Calculate the maximum data volume, in MB, that can be transmitted in five minutes at a communication speed of 10 Mbps.

(4) Calculate the time required to transfer 20 MB of data at a communication speed of 80 Mbps. Assume that the transfer efficiency is 100% and disregard any data other than the data volume.

(5) Calculate the time required to transfer 1 GB of data at a communication speed of 20 Mbps. Assume that the transfer efficiency is 80% and disregard any data other than the data volume.

Exercise

1 Answer the following questions. In these questions, assume that 1 KB = 1,000 B, 1 MB = 1,000 KB, and 1 GB = 1,000 MB.

(1) Bps is a unit used to measure data communication speed. Choose the appropriate explanation for 40 Mbps from the options **A** to **D** below, and answer using the letter.
- **A** Capable of transferring data at 40 megabits per second.
- **B** Capable of transferring data at 40 megabytes per second.
- **C** Capable of transferring data at 40 megabits per minute.
- **D** Capable of transferring data at 40 megabytes per minute.

(2) Calculate the maximum data volume, in bytes, that can be transmitted in one minute at a communication speed of 120 bps?

(3) Calculate the data volume, in MB, that can be transmitted in three minutes at a communication speed of 20 Mbps.

(4) Calculate the time required to transfer 15 MB of data at a communication speed of 60 Mbps. Assume that the transfer efficiency is 100% and disregard any data other than the data volume.

(5) Calculate the time required to transfer 1 GB of data at a communication speed of 25 Mbps. Assume that the transfer efficiency is 80% and disregard any data other than the data volume.